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	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: A study is underway to conduct an experimental and theoretical research program on the effect of an external radiant flux on the ignition, subsequent flame spread, and fire suppression of a representative combustible material in space exploration atmospheres (SEA). The near-term objective of the research is to develop a ground based Material Ignition and Suppression Test (MIST) apparatus that reflects the environments that are expected in space-based facilities and is similar to the Solid Fuel Ignition and Extinction / Material Ignition and Suppression Test (SoFIE/MIST) apparatus that will be installed in the International Space Station (ISS). The ground based MIST apparatus has already been built and tested during the research program for NASA Grant #NNX10AE01G, and the apparatus is currently being used to obtain a flammability boundary, ignition, flame spread and extinction, of a polymeric material, specifically PMMA, which can be viewed as a baseline material for fire testing. As the microgravity data from the SoFIE/MIST		

Task Description:	experiments is available, it will be analyzed and compared with the normal gravity data to determine the effect of reduced gravity on the flammability of materials in SEA. Methods/techniques: The experimental apparatus consists of a small-scale rectangular flow duct that has a cylindrical fuel sample at the axis of the duct and has four radiant heaters placed in the corners of the duct that radiate the sample surface with a predetermined radiant flux. An electrical coil ignitor is placed at the trailing edge of the sample to ignite the sample and initiate flame spread. For SEA testing in normal gravity, the apparatus is placed inside a pressure chamber that resembles that of the Combustion Integrated Rack (CIR) in the ISS. An oxidizer gas, at SEA oxygen concentrations and pressures, flows at low velocity through the length of the duct. The fuel samples are rods of black PMMA of different diameters. The radiant heaters provide a radiant flux over the sample surface that increases its temperature and enhances the rate of flame spread. The ignitor provides the necessary energy to ignite the polymeric samples. The tests provide information about the effect of a radiant flux on the ignition, flame spread, and burning rate at a given oxygen concentration and ambient pressure. This will be followed by the reduction of the O2 concentration with N2 or CO2 until extinction occurs. The result will provide information about the suppression characteristics of inert gases in SEA and of the effect of SEA on the Limiting Oxygen Index (LOI) of the material when exposed to an external radiant flux. The experimental task will be complemented with a theoretical task that includes both analytical and numerical modeling. The analytical modeling uses simplified formulas of flame spread previously developed that incorporate the effect of the external radiant flux on the flame spread rate through the energy received by the fuel rod at a particular time. The numerical model will use a solid combustible pyrolysis numerical code developed in the Principal Investigator (PI) laboratory, together with a computational fluid dynamics (CFD) code simulating the reacting gas phase. The microgravity experiments conducted under the SoFIE project will be used to validate the theoretical model. It is the objective that the theoretical model will prevent the need to conduct extensive experimental testing of materials in microgravity. Perceived Significance: There is insufficient knowledge about fire behavior of materials in environments expected in space-based facilities, i.e., SEA (micro/partial-gravity, low velocity flow, reduced pressure, and elevated oxygen concentrations). Nor is there a testing methodology specifically developed to determine the fire hazards of materials under those conditions. It is the long-term objective of this project to provide additional information about the effect of SEA on the flammability (ignition, flame spread, and heat release rate) and suppression of materials, and guidance to interpret and extend current NASA testing procedures to environments expected in space-based facilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, burning (flame spread and mass burning), and extinction characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
	This progress report presents a summary of the research conducted during the last year of the current funding period. The research conducted in previous years has been already reported in the corresponding annual research reports. It includes a brief description of the normal gravity MIST experimental apparatus and the results obtained during this period. More detailed information is provided in the annual report for this grant period. [Ed. Note: Technical reports are delivered to NASA on an annual basis. For annual MIST reporting prior to FY2022, see Grant #NNX13AL10A.] Experimental Apparatus The experimental apparatus used in the normal gravity experiments has been described in previous progress reports and thus it will be described here only briefly. It consists of a laboratory scale combustion tunnel that is inserted in a 105 L pressure chamber. The flow duct of the combustion tunnel was built following a similar design of the apparatus that will be used in the Solid Fuel Ignition and Extinction / Material Ignition and Suppression Test (SoFIE/MIST) microgravity experiments that are planned to be conducted aboard the International Space Station (ISS). The laboratory scale combustion tunnel has a cross sectional area of 125 mm by 125 mm, and is 600 mm in length. The first 350 mm of the duct serves as a flow straightener, where inlet gases pass sequentially through a 20 mm layer of 5 mm borosilicate beads, a stainless-steel mesh, and 63 mm thick aluminum honeycomb with 3 mm cells before entering the test section. The last 250 mm portion of the duct is used as the test section. The side walls of the top portion of the test section are made of Pyrex glass to withstand the heat from the heater array and the burning samples, while allowing the visualization of the burning samples. The pressure chamber is provided with a flow system that provides constant supply, and exhaust, of gases to avoid vitiation problems. Compressed house air (or nitrogen and oxygen) is supplied through critical nozzles (O'Keefe Controls) to the bottom of the combustion tunnel while constantly evacuating to maintain constant the pressure inside the chamber. The chamber pressure is controlled with a high-capacity vacuum generator (Vaccon JS-300) and a mechanical vacuum regulator. The chamber pressure is monitored constantly with an electronic pressure transducer (Omega Engineering, Inc. PX303-015A5V). Tests can be conducted at sub-atmospheric pressures ranging between 100 kPA and 30 kPa ± 2 kPa, and varied oxygen concentrations. The forced flow velocity can be varied up to 30 cm/s. The samples used in the experiments are cylindrical rods made of cast black polymethyl methacrylate (PMMA). The rods are placed vertically in the center of the test section, supported on a metal cylinder of the same diameter to prevent flow disturbances. The black PMMA was selected to maximize the absorptance of the thermal radiation at the rod's surface. The PMMA rods are 90 mm long with diameters of 6.4 mm, 9.5, mm and 12.7 mm. The sample material and geometry were selected following previous experiments conducted in the ISS under the Burning and Suppression of Solids - II (BASS-II) project and the experiments that will be done as part of the SoFIE/MIST project. Also, PMMA is potentially going to be used in some spacecraft components, such as windows, in future space vehicles. Several K-type thermocouples are embedded on the surface of the PMMA samples to record surface temperature from the beginning of the test until the moment the flame arrives to the location of the bead. Four flat ceramic heaters (Bach Resistor Ceramics GmbH) are used as the source of external thermal radiation. These heaters were selected to match the heaters that will be used in the MIST/SoFIE experiments. Each ceramic heater has a length of 75 mm, a width of 14.7 mm, and a heated section of 50 mm. The four heaters are mounted on fixed rails positioned on each one of the corners of the experimental section surrounding the sample The heaters are positioned 83 mm from the top opening of the duct, aligned with the

Task Progress:	position of the sample. The four heaters are used to provide a variable heat flux in the direction of flame spread (downward). The distribution of heat flux over the solid surface provided by the heaters was characterized under different power settings using a Schmidt-Boelter radiometer from Medtherm Corporation. The characterization of the heaters was performed considering the heaters assembled in the tunnel with the array of reflectors. Experiment The experimental research conducted during this reporting period consists of a study of the effect of the ambient conditions on the spread of flames and limits of extinction of a solid combustible. We are specifically interested in determining the effect of sub-atmospheric pressure and external heating on the flame spread rate of PMMA rods and their extinction limits (Limited Oxygen Concentration, or LOC). Sub-atmospheric pressures are present in various environments such as spacecraft and space habitats with space exploration atmospheres (SEA), aircraft, and high-altitude locations. External radiant heating is included to simulate the effect of heating from a nearby fire or overheated equipment. This aspect of a fire is tested in the LIFT (Lateral Ignition and Flame Spread Test) apparatus - ASTM E-132 to rank the flammability of materials according to their ignition and flame spread characteristics. In the LIFT apparatus, the samples are exposed to a decaying radiant flux from the radiant heater placed. The data are used to generate a flammability diagram for the combustible material based on ignition delay and flame spread rate. The determination of the flame spread extinction requirements is done following the methodology of the Limiting Oxygen Index (LOI) apparatus - ASTM 2863; that determines the limiting concentration of oxygen for flame spread. The LOI test uses a vertical cylindrical rod of the material to be tested placed in a vertical cylindrical tube in diameter. A forced flow of controlled oxygen concentration of 5 cm/s is flown upward in the tube past the rod. The sample is ignited on the top of the cylinder such that flames spread downward opposing a mixed (free and forced) flow. The oxygen concentration is reduced slowly as the sample burns, until the flame stops spreading and is extinguished. The test apparatus used in the present project combines the LOI and Forced Flow Ignition and Flame Spread Test (FIST) tests into one, to determine the Limiting Oxygen Concentration (LOC) of vertical PMMA rods in a low velocity opposed flame spread and an incident decreasing radiant flux, in sub-atmospheric pressures. In addition to providing information about the flammability of materials in Earth atmospheres, the results of the tests also provide data for comparison with experiments to be conducted in the ISS under the SoFIE-MIST. The comparison between LOC data in normal and microgravity would provide further understanding of the LOC of materials in SEA, and, in turn, information about the fire safety in future spacecraft and space facilities. Experiments were conducted for 40 kPa to 100 kPa ambient pressures with intervals of 10 kPa, and at external heat fluxes of 0 kW/m2, 0.9 kW/m2, 1.9 kW/m2, and 3.3 kW/m2. The LOC is difficult to identify; as the LOC is approached, any minor perturbation can be enough to extinguish the flame. To reduce the error, the oxygen concentration was reduced in small intervals and the tests were repeated between three and five times. It was found that the average of the oxygen volume fraction at which the flame extinguished (LOC) is a function of the chamber pressure and of the different heat fluxes. It was observed that for all the radiant fluxes, reducing the ambient pressure increased the LOC, although not strongly. This is due to the deterrent effect of ambient pressure on the flame spread rate and indicates that materials are less flammable as the ambient pressure is reduced. Increasing the external radiant flux results in a lower LOC. This is because the added energy preheats the sample ahead of the flame front, and therefore increases the flame spread rate. It was found that the LOC data can be approximately correlated in terms of the partial pressure of oxygen. The data for each heat flux are almost linearly related, with a reduction of the partial pressure of oxygen, and a slight reduction in the slope for higher heat fluxes. Since, for a given total ambient pressure, solid fuels are flammable for partial pressure of oxygen above the LOC, it is found that PMMA is flammable for partial pressures of oxygen below the oxygen partial pressure of air. Furthermore, in SEA air environments, PMMA is flammable for pressures smaller than approximately 110 kPa. The presence of an external heat flux decreases the LOC partial pressure of oxygen even further. Determining the partial pressure of oxygen in the context of flammability limits is important because it provides an objective basis for comparing experiments conducted in various environmental conditions, as it combines the oxygen concentration and the ambient pressure into a single term. Analysis The finding that the LOC of a combustible solid is given by the oxygen mole fraction is somewhat surprising, although phenomenological arguments can be used to support it. The simplified analysis of flame spread developed by the Principal Investigator (PI) of the project is a possible approach to discuss this issue. In that work, an expression was derived to describe the flame spread over a solid fuel based on the concept that the flame spreads over the solid as a series of ignition steps, where the flame acts both as a heat source for fuel pyrolysis, and as an ignition pilot (the flame) of the pyrolyzate and oxidizer gas ahead of the flame. The analysis provides a simplified analytical formula for the opposed flame spread rate obtained from the ratio of the solid heating length ahead of the pyrolysis front and the time required to heat, pyrolyze, and ignite the solid. The equation contains two terms, one that describes the heat transfer mechanisms of the flame spread over a thin rod and another term that represents the gas phase chemical kinetics mechanisms. For fast reaction rate, the chemical kinetics term is small, and it does not play a role in the flame spread rate. In this case, only heat transfer determines the flame spread rate. The resulting equation normally is referred to as describing the heat transfer regime of flame spread. As the reaction term decreases, the gas induction time increases, and consequently the rate of flame spread also decreases. The chemical kinetics also affects the heat transfer term through the flame temperature. Under these conditions, both terms in the equation play a role in the flame spread process and must be considered. If the reaction rate decreases even more, the flame extinguishes. From the phenomenological analysis, it can be concluded that the oxygen mole fraction is a primary determining parameter of the LOC for flame spread, which supports the observed experiments.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
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